

Project 4: Circuit Element Identification

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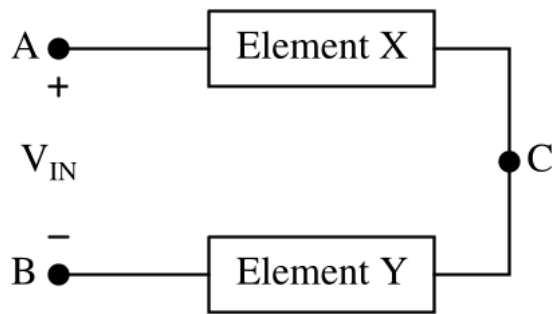
EEL3123-C0013: Linear Circuits II

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4.A Objective



Node A is connected to a red wire.

Node B is connected to a black wire.

Node C is connected to a green wire.

Box#4

The objective of Project 4 was to determine. The unknown load could be a purely resistive, RL, or RC network.

Equipment:

- Concealed AC load box (Unknown RR, RL, or RC)
- Digital multimeter (DMM)
- Reference resistor
- Oscilloscope
- USB storage

4.B Plan

A known reference resistor is connected in series with the concealed box so that the circuit current can be calculated from a voltage measurement.

1. Measure the RMS voltages across the source, reference resistor, and each concealed load.
2. Use the reference-resistor voltage to calculate circuit RMS current. $I = I_R = \frac{V_R}{R_R}$
3. Measure the phase difference between V_X and V_R with the oscilloscope.
4. Measure the phase difference between V_Y and V_R with the oscilloscope.
5. Determine the types of components for Element X and Element Y.
6. Calculate the resistance of the unknown resistor $R_P = \frac{V_P}{I_{rms}}$.
(We will call this Element P. It could be Element X or Element Y)
7. Calculate the complex impedance of the other component $Z_Q = \frac{V_Q}{I_{rms}}$.
(We will call this Element Q. It could be Element X or Element Y)
8. Calculate the value of Element Q.
 - If $\text{phs}_Q \approx 0^\circ$ Element Q is a resistor and $Z_Q = R_Q$
 - If $\text{phs}_Q > 0^\circ$ Element Q is an inductor and $Z_Q = j\omega \frac{L_Q}{R_L}$
 - If $\text{phs}_Q < 0^\circ$ Element Q is a capacitor and $Z_Q = \frac{1}{j\omega C_Q}$

4.C Results

With $V_{IN} = 2V_{PP}@1kHz$, we measured the following RMS voltages with our DMM:

$$V_{IN} = 1.42V$$

$$V_R = 233mV$$

$$V_X = 1.02V$$

$$V_Y = 104mV$$

Eq 1

This gives us an RMS current of:

$$\begin{aligned}
 I_{rms} &= \frac{V_R}{R_R} \\
 &= \frac{0.233V}{980.2\Omega} \\
 &= 238\mu A
 \end{aligned}
 \tag{Eq 2}$$

Next, we determine that the phase difference between V_X and V_R is 0° .
And, the phase difference between V_Y and V_R is 15.48° leading.

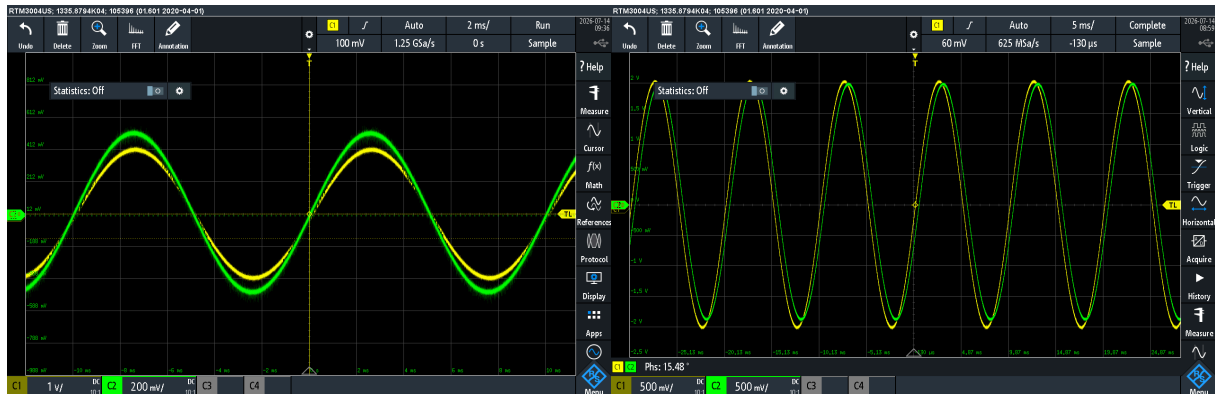


Figure 2:

Left: Phase difference between V_X and V_R
Right: Phase difference between V_Y and V_R

This leads us to our conclusion for the internal layout of the box (Figure 3).

We can now calculate the resistance of Element P (X) and the complex impedance of Element Q (Y).

$$\begin{aligned}
 R_P &= \frac{V_P}{I_{rms}} \\
 &= \frac{1.02V}{238\mu A} \\
 &= 4.29k\Omega
 \end{aligned}
 \tag{Eq 3}$$

$$\begin{aligned}
 Z_Q &= \frac{V_Q}{I_{rms}} \\
 &= \frac{0.104V}{238\mu A} \\
 &= 2.119k\Omega
 \end{aligned}
 \tag{Eq 4}$$

Finally we use the impedance of Element Q (Equation 4) to determine the value of Element Q which we have already identified as a capacitor.

$$\begin{aligned}
 Z_Q &= \frac{1}{j\omega C_Q} \\
 C_Q &= \frac{1}{j\omega Z_Q} \\
 &= \frac{1}{j \times 2\pi \times 1k \text{ Hz} \times 2.119k\Omega} \\
 &= 80nF
 \end{aligned}
 \tag{Eq 5}$$

This gives us our final results for the concealed box:

$$\begin{aligned}
 &\text{Element X, Resistor, } 4.29k\Omega \\
 &\text{Element Y, Capacitor, } 80nF
 \end{aligned}
 \tag{Eq 6}$$

4.D Conclusion

After crossreferencing our results with the available components in the lab, we determined that the concealed box contained a $4.6k\Omega$ resistor and a $100nF$ capacitor. The discrepancy between our calculated values and the actual values can be attributed to measurement error and component tolerances. The final circuit is shown below (Figure 3).

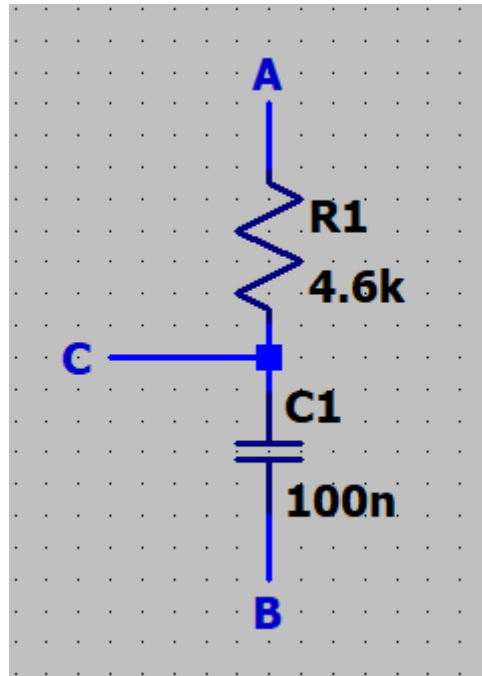


Figure 3: The internal circuit of the concealed box